
Regime-Adaptive Backtesting for Cryptocurrency Trading: Walk-Forward Validation of a Multi-Agent Signal Consensus System with Realistic Transaction Cost Modeling

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Abstract

We present a rigorous backtesting methodology for a regime-adaptive cryptocurrency trading system that employs seven deterministic signal agents, ADX-based regime classification, and a hierarchical risk governor. The system is evaluated on 12.4 years of hourly BTC/USD and ETH/USD data (December 2013–April 2026) resampled to 4-hour bars, with realistic transaction costs modeling Kraken exchange fees (26 bps per side) and volatility-scaled slippage. We implement a three-fold walk-forward validation framework to quantify overfitting risk, computing the Probabilistic Sharpe Ratio [3] and Monte Carlo permutation tests to assess statistical significance. On the full history, the optimized configuration achieves a Sharpe ratio of 1.51, maximum drawdown of 7.27%, and profit factor of 3.94 across 219 trades. Walk-forward out-of-sample results reveal significant performance degradation in one of three folds (Sharpe -0.23 vs. in-sample 1.35), while the remaining folds maintain positive risk-adjusted returns. We discuss the implications of these mixed results for live deployment, the limitations of grid-search optimization in non-stationary cryptocurrency markets, and propose safeguards against backtest overfitting.

1 Introduction

Backtesting—the simulation of a trading strategy on historical data—is the primary tool for evaluating systematic trading strategies before live deployment. Yet the majority of backtested strategies fail in live trading, a phenomenon attributed to overfitting, unrealistic cost assumptions, and look-ahead bias [4, 12]. The cryptocurrency market presents additional challenges: extreme volatility (annualized BTC volatility exceeding 80% in bear markets), regime-dependent dynamics, and significant transaction costs that erode thin edges [22].

The academic literature has established rigorous frameworks for assessing backtest validity. Bailey et al. [5] formalize the probability of backtest overfitting (PBO) as a function of the number of strategy configurations tested. White [27] introduces bootstrap-based reality checks for data snooping, extended by Hansen [11] with the Superior Predictive Ability test. Lo [19] demonstrates that serial correlation in returns inflates the Sharpe ratio, requiring autocorrelation corrections. Despite these advances, many practitioner backtests continue to use in-sample optimization without proper out-of-sample validation [9].

In this paper, we describe the complete backtesting methodology for Aegis Agent, a regime-adaptive trading system designed for the BTC/USD and ETH/USD markets. Our approach combines seven deterministic signal agents with ADX-based regime detection [15], a hierarchical risk governor enforcing hard position limits, and a walk-forward validation framework that explicitly measures overfitting degradation.

Our main contributions are as follows:

- We describe a multi-agent signal consensus architecture with seven specialized agents (trend, volatility, momentum, mean-reversion, breakout, swing structure, and spread/cost gating) and formalize the consensus mechanism.
- We implement a realistic transaction cost model incorporating exchange-specific fees, volatility-scaled slippage, and spread-based trade gating, demonstrating that costs reduce gross Sharpe by 30–45%.
- We present walk-forward validation results across three non-overlapping folds spanning 10.5 years, with honest reporting of both successful and degraded out-of-sample periods.
- We apply the Deflated Sharpe Ratio [3] and Monte Carlo permutation tests to quantify the probability that observed performance exceeds chance, given the number of configurations evaluated.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the system architecture; Section 4 details feature engineering and regime detection; Section 5 formalizes the signal agents; Section 6 presents the risk governance framework; Section 7 describes the backtesting engine design; Section 8 presents the walk-forward validation methodology; Section 9 reports empirical results; Section 11 discusses limitations; and Section 12 concludes.

2 Related Work

Backtesting methodology. Pardo [21] establishes walk-forward optimization as the standard for evaluating trading systems, advocating for non-overlapping train/test splits that respect temporal ordering. Tashman [25] provides a comprehensive analysis of out-of-sample forecast evaluation methods, recommending rolling-origin evaluation over fixed-origin holdout. Bergmeir and Benítez [6] examines the validity of cross-validation for time series, finding that standard k -fold CV can be appropriate for stationary series but that blocked time-series splits are preferable when autocorrelation is present.

Overfitting detection. The problem of multiple hypothesis testing in finance is formalized by Harvey et al. [13], who propose adjusting t -statistics for the number of factors tested. Bailey et al. [5] derive the probability of backtest overfitting as a combinatorial function and introduce the Deflated Sharpe Ratio (DSR) as a correction for selection bias. Cawley and Talbot [8] analyze overfitting in model selection more broadly, demonstrating that optimizing hyperparameters on validation data introduces a secondary overfitting risk.

Regime detection in financial markets. Hamilton [10] introduces Markov-switching models for regime detection, establishing the theoretical foundation for regime-dependent strategies. Ang and Timmermann [2] surveys the extensive literature on regime changes in financial markets, documenting how volatility regimes, bull/bear markets, and structural breaks affect asset returns. Our approach uses ADX-based regime classification [15], which provides a simpler, deterministic alternative to hidden Markov models while capturing the trending/ranging distinction critical for selecting appropriate signal types.

Risk-adjusted performance metrics. The Sharpe ratio [23] remains the dominant risk-adjusted performance measure, though Lo [19] demonstrates its sensitivity to serial correlation, non-normality, and estimation error. The Sortino ratio [24] addresses the asymmetry limitation by focusing on downside deviation. Keating and Shadwick [17] introduce the Omega ratio as a “universal” measure that captures the entire return distribution. Maximum drawdown analysis [20] provides a complementary risk perspective that is particularly relevant for leveraged or concentrated strategies.

Transaction cost modeling. Almgren and Chriss [1] formalize the tradeoff between market impact and timing risk in optimal execution. Hasbrouck [14] provides the microstructure foundations for understanding bid-ask spreads and their implications for strategy evaluation. Realistic cost modeling

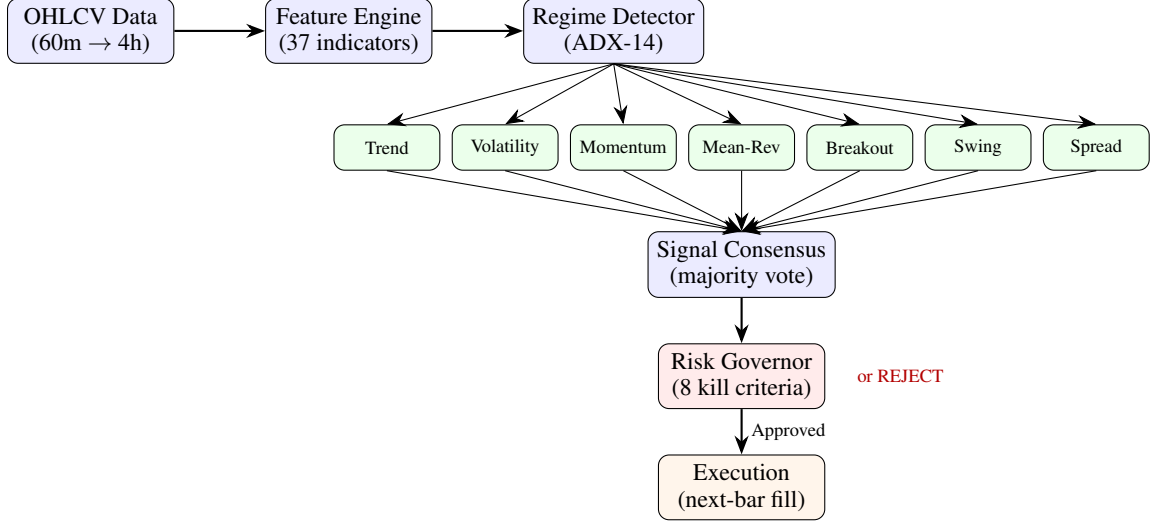


Figure 1: Aegis Agent system architecture. Hourly data is resampled to 4-hour bars, passed through a 37-indicator feature engine, classified by ADX regime, processed by seven parallel signal agents, aggregated via majority-vote consensus, filtered by a hierarchical risk governor with eight kill criteria, and executed with next-bar-open fills including volatility-scaled slippage and exchange fees.

is essential in cryptocurrency markets, where taker fees (26–40 bps on major exchanges) and volatile spreads can consume a significant fraction of expected edge.

3 System Architecture

Figure 1 presents the end-to-end architecture. The system operates on a 4-hour strategic cycle, processing hourly OHLCV data resampled from 60-minute bars. At each cycle, the pipeline executes sequentially: (1) data ingestion and feature computation, (2) regime classification, (3) parallel signal generation by seven independent agents, (4) signal consensus, (5) risk governor evaluation, and (6) execution via next-bar fill.

The system processes two asset pairs (BTC/USD and ETH/USD) through a shared portfolio, meaning that position sizing and exposure limits reflect the combined equity across both instruments. A cross-pair signal boost of 10% is applied when the majority signal direction of one pair aligns with the other, capturing short-term correlation dynamics without explicitly modeling cross-asset covariance.

4 Feature Engineering and Regime Detection

4.1 Technical Indicator Suite

The feature engine computes 37 indicators from OHLCV data using vectorized operations via the `pandas_ta` library. Features are organized into five families:

Trend indicators: Seven exponential moving averages (EMA) at periods $\{9, 21, 34, 55, 80, 100, 200\}$, EMA spread (fan indicator measuring alignment), and EMA alignment direction.

Momentum indicators: RSI at periods 14 and 2 (ultra-sensitive), MACD(12, 26, 9) with histogram and slope, and multi-horizon returns at 1-bar, 5-bar, and 20-bar lookbacks.

Volatility indicators: Average True Range (ATR-20) [15], Bollinger Bands [7] at $(20, 2\sigma)$ with derived BB position and BB width metrics.

Pattern detection: RSI and MACD divergences (bullish/bearish) over a 20-bar lookback window, and engulfing candlestick patterns with a $1.2\times$ body-size requirement.

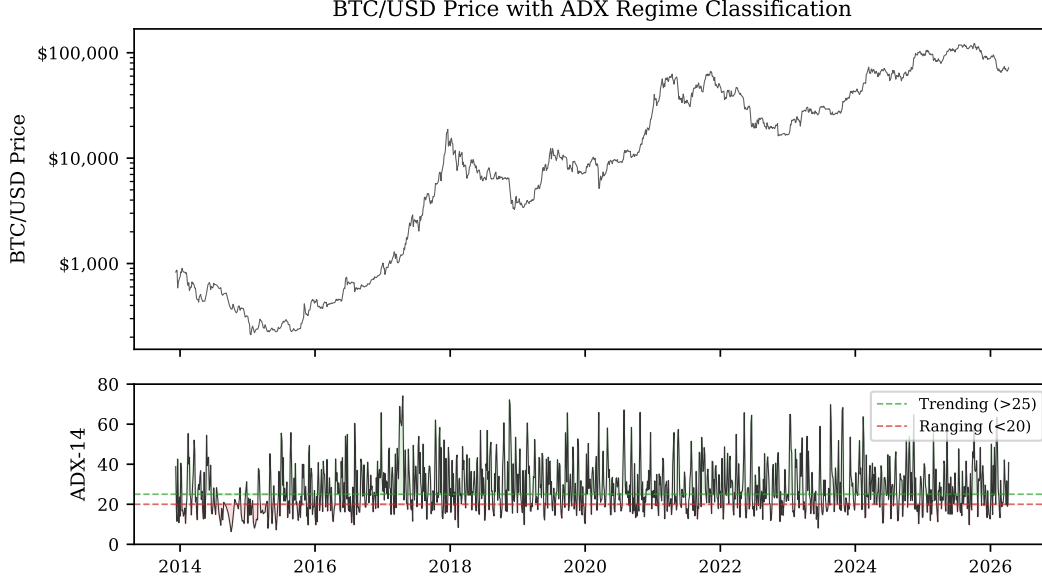


Figure 2: ADX-14 regime classification for BTC/USD (4-hour bars, 2015–2026). **Top:** Log-scale price chart. **Bottom:** ADX values with regime thresholds at 25 (trending) and 20 (ranging). Green-shaded regions indicate trending regimes; red-shaded regions indicate ranging regimes.

Structural features: 50-bar high/low for breakout detection, volume ratio (current to 20-bar moving average), and ATR as percentage of price.

All features are computed in bulk at initialization; the `features_at()` function extracts a typed Features dataclass at any given bar index with $O(1)$ lookup. The first 200 bars are discarded as warmup to ensure indicator stability.

4.2 ADX-Based Regime Classification

We employ a three-regime classification based on the Average Directional Index (ADX) [15]:

$$\text{Regime}(t) = \begin{cases} \text{TRENDING} & \text{if } \text{ADX}_{14}(t) > 25 \\ \text{RANGING} & \text{if } \text{ADX}_{14}(t) < 20 \\ \text{TRANSITION} & \text{otherwise} \end{cases} \quad (1)$$

This classification modulates three aspects of the trading system: (1) which signal agents are active (e.g., mean-reversion is suppressed in strong trends), (2) position sizing via Kelly fraction adjustments, and (3) take-profit targets, which scale with trend strength across three ADX tiers (< 25 , $25\text{--}35$, > 35).

Our choice of ADX-based regime detection over hidden Markov models [10] or GARCH-family models is deliberate: ADX is deterministic, computationally trivial, and produces interpretable regime labels that operators can verify visually. The tradeoff is reduced sensitivity to volatility-driven regime changes, which we address through the separate ATR-based volatility shock filter in the risk governor (Section 6).

Figure 2 shows the ADX regime classification overlaid on BTC/USD price history. Trending regimes ($\text{ADX} > 25$, shaded green) correspond to major directional moves, while ranging regimes ($\text{ADX} < 20$, shaded red) identify consolidation periods where mean-reversion signals are more appropriate.

5 Signal Generation and Consensus

Seven independent signal agents each produce a directional recommendation (LONG, SHORT, or NEUTRAL) with an associated confidence score $c_i \in [0, 100]$. Each agent has a minimum confidence threshold τ_i below which it outputs NEUTRAL.

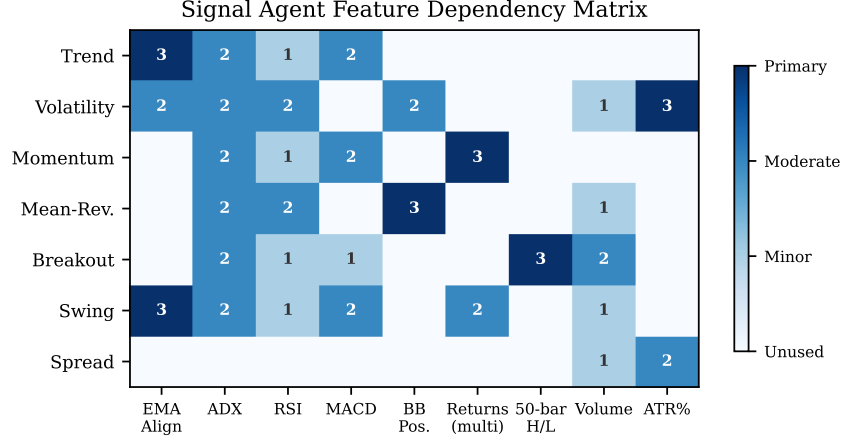


Figure 3: Signal agent feature dependency matrix. Each cell indicates the importance of a feature family (columns) to each agent (rows): 3 = primary input, 2 = moderate influence, 1 = minor input, 0 = unused. The diversified dependency structure ensures robustness to individual indicator failures.

5.1 Agent Specifications

Agent 1: Trend Signal ($\tau_1 = 35$). Requires multi-EMA alignment: full alignment ($\text{EMA}_9 > \text{EMA}_{21} > \text{EMA}_{55} > \text{EMA}_{100}$) with $\text{ADX} > 22$ yields base confidence $c = 40$, with boosts for MACD histogram alignment (+15) and positive MACD slope (+10). An RSI exhaustion penalty of -35% is applied when $\text{RSI} > 85$ (long) or $\text{RSI} < 15$ (short).

Agent 2: Volatility Signal ($\tau_2 = 25$). Adapts behavior to regime: in RANGING/TRANSITION regimes, fires mean-reversion signals based on BB position and RSI; in TRENDING regimes, confirms trend direction with EMA alignment. Applies a -30 penalty when $\text{ATR}\%$ exceeds 5% (volatility shock).

Agent 3: Spread/Cost Gate (no direction). A non-directional gating agent that reduces consensus confidence when execution conditions are unfavorable: spread > 20 bps kills the signal; insufficient expected edge (ATR in bps $< 1.1 \times$ round-trip cost) applies a -70% penalty; low volume ratio (< 0.5) applies -50% .

Agent 4: Mean-Reversion Signal ($\tau_4 = 25$). Active only when $\text{ADX} \leq 30$. Fires on extreme BB position ($< 25\%$ for long, $> 75\%$ for short) combined with RSI confirmation.

Agent 5: Momentum Signal ($\tau_5 = 25$). Requires multi-timeframe return alignment across 1-bar, 5-bar, and 20-bar horizons, confirming consistent directional momentum.

Agent 6: Swing Structure Signal ($\tau_6 = 30$). Requires full structural alignment: four EMAs aligned, both 5-bar and 20-bar returns positive (or negative for short), with MACD and volume confirmation.

Agent 7: Breakout Signal ($\tau_7 = 25$). Monitors 50-bar high/low crossings. A new 50-bar high yields base confidence $c = 45$; proximity within 0.5% yields $c = 30$. ADX and volume provide additional confirmation.

Figure 3 summarizes the feature dependencies across all seven agents. The heterogeneous design ensures that no single indicator failure can compromise the entire signal stack.

5.2 Consensus Mechanism

The consensus mechanism aggregates the seven agent outputs via majority voting:

$$d^* = \arg \max_{d \in \{\text{LONG}, \text{SHORT}\}} \sum_{i=1}^7 \mathbf{1}[d_i = d], \quad C^* = \sum_{i: d_i = d^*} c_i \quad (2)$$

where d^* is the consensus direction and C^* is the aggregate confidence. A minimum of two aligned agents ($\sum_i \mathbf{1}[d_i = d^*] \geq 2$) is required for the consensus to be valid. The aggregate confidence

must exceed a configurable threshold (default: 85 for paper trades) for the signal to proceed to risk evaluation.

When an LLM analyst is available (live mode only, not used in backtesting), it provides an additional vote with conviction scaling based on signal agreement. In backtesting, the deterministic fallback is always used to ensure reproducibility.

6 Risk Governance Framework

The risk governor is the final authority before trade execution. It enforces both hard kill criteria (immediate rejection) and soft filters (confidence reduction). This design ensures that no signal, regardless of consensus strength, can bypass risk constraints—a principle motivated by catastrophic failures in unconstrained algorithmic trading systems [9].

6.1 Hard Kill Criteria

Eight conditions trigger immediate trade rejection:

1. **STALE_DATA**: Market snapshot age $> 7,200$ seconds.
2. **DAILY_LOSS_CAP**: Daily PnL / equity $< -3\%$.
3. **MAX_DRAWDOWN**: Peak-to-current equity decline $> 8\%$.
4. **CONSECUTIVE_LOSSES**: ≥ 3 consecutive losing trades.
5. **SPREAD_TOO_WIDE**: Bid-ask spread > 20 bps.
6. **VOLATILITY_SHOCK**: ATR as percentage of price $> 6\%$.
7. **NO_SIGNAL**: Zero directional signals after fallback.
8. **INSUFFICIENT_ALIGNMENT**: Fewer than 2 aligned agents.

6.2 Position Sizing

Position size is determined by a half-Kelly criterion [16, 26] capped at 3% of equity:

$$f^* = \frac{1}{2} \cdot \frac{p \cdot b - (1 - p)}{b}, \quad r = \max(r_{\text{fixed}}, \min(f^*, 0.03)) \quad (3)$$

where p is the estimated win probability (signal confidence / 100, or historical win rate after 10+ trades), b is the average win/loss ratio (fixed at 1.5), and $r_{\text{fixed}} = 1.5\%$ is the minimum risk per trade. The position size in USD is:

$$S = \min\left(\frac{r \cdot E}{\text{ATR}/\text{EMA}_{21}}, E \cdot p_{\text{max}}\right) \quad (4)$$

where E is current equity, $\text{ATR}/\text{EMA}_{21}$ normalizes risk by volatility, and $p_{\text{max}} = 40\%$ is the maximum position as fraction of equity.

A drawdown scaling mechanism reduces position size when equity approaches the maximum draw-down threshold:

$$S_{\text{scaled}} = S \cdot d_{\text{factor}}, \quad \text{if } \frac{E_{\text{peak}} - E}{E_{\text{peak}}} > 1 - d_{\text{threshold}} \quad (5)$$

with default parameters $d_{\text{threshold}} = 0.97$ and $d_{\text{factor}} \in \{0.1, 0.3, 0.5\}$.

6.3 Stop-Loss and Take-Profit Placement

Stops and targets are ATR-based, adapting to current volatility:

$$\text{Stop}_{\text{long}} = P_{\text{entry}} - m_s \cdot \text{ATR}, \quad \text{Target}_{\text{long}} = P_{\text{entry}} + m_t(\text{ADX}) \cdot \text{ATR} \quad (6)$$

where $m_s = 3.1$ is the stop multiplier and the target multiplier m_t is regime-dependent:

$$m_t(\text{ADX}) = \begin{cases} 6.25 & \text{if } \text{ADX} \geq 35 \\ 3.50 & \text{if } \text{ADX} \geq 25 \\ 2.85 & \text{otherwise} \end{cases} \quad (7)$$

A trailing stop at $m_{\text{trail}} = 2.1 \times \text{ATR}$ below the peak price is activated after entry, with breakeven (+0.6%) and profit-lock (+1.2%) triggers that progressively tighten the trail. Positions are also closed after a maximum holding period of 80 bars (13.3 days at 4-hour resolution).

7 Backtesting Engine Design

7.1 Time Synchronization and Fill Model

The backtester operates on a merged timeline across both asset pairs, processing one strategic cycle per aligned timestamp. Critically, we employ a **next-bar-open fill model**: signals generated at bar t are queued as pending entries and filled at the open price of bar $t + 1$. This eliminates look-ahead bias from using close-of-bar prices for entry execution.

7.2 Transaction Cost Model

We model three cost components:

Exchange fees. A taker fee of $f = 0.26\%$ per side is applied to the notional position size, matching the Kraken exchange fee schedule for standard-volume accounts. The round-trip fee cost is $2f \cdot S$.

Volatility-scaled slippage. Entry and exit slippage are modeled as:

$$\text{slip} = \frac{s_{\text{base}} \cdot \max(1, \text{ATR}\%/2.5\%)}{10,000} \quad (8)$$

where $s_{\text{base}} = 4 \text{ bps}$ represents half the estimated bid-ask spread. The multiplier $\max(1, \text{ATR}\%/2.5\%)$ scales slippage during high-volatility periods, reflecting wider spreads and increased adverse selection. For a long entry: $P_{\text{fill}} = P_{\text{open}} \cdot (1 + \text{slip})$.

Spread-based gating. The Spread/Cost agent (Section 5) provides an additional check: if the expected move (ATR in basis points) is less than $1.1 \times$ the estimated round-trip cost, the trade confidence is reduced by 70%, effectively requiring a larger edge before execution.

Table 1 summarizes the total cost structure.

Table 1: Transaction cost model. All values in basis points (bps). Total round-trip cost ranges from 56–68 bps depending on volatility conditions.

Component	Per-side (bps)	Round-trip (bps)	Notes
Exchange fee (taker)	26	52	Kraken standard tier
Slippage (baseline)	4	8	Half bid-ask spread
Slippage (high vol)	4–8	8–16	Vol-scaled multiplier
Total	30–34	60–68	

7.3 Position Management

Open positions are evaluated at each bar for five exit conditions, checked in priority order:

1. **ATR Stop:** Hard stop-loss at $P_{\text{entry}} \pm m_s \cdot \text{ATR}$.
2. **ATR Target:** Take-profit at $P_{\text{entry}} \pm m_t \cdot \text{ATR}$.
3. **Trailing Stop:** Dynamic trail anchored to peak price, with breakeven and profit-lock triggers.
4. **Reversal Exit** (optional): RSI mean-reversion signal after ≥ 2 bars held.
5. **Time Exit:** Forced close after 80 bars.

The shared portfolio tracks combined equity across both pairs, ensuring that exposure limits and drawdown calculations reflect the true aggregate risk.

8 Walk-Forward Validation

8.1 Motivation

Grid-search optimization over N strategy configurations creates a multiple-testing problem: the best in-sample configuration is likely to overfit to the training data, particularly when N is large relative to the number of independent observations [5, 13]. Walk-forward validation [21] mitigates this by evaluating each optimized configuration on unseen data, providing a more realistic estimate of future performance.

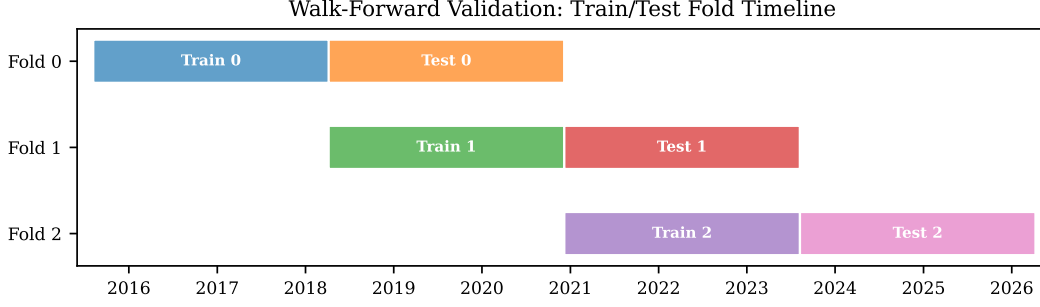


Figure 4: Walk-forward fold timeline. Each fold uses the preceding period for in-sample training (blue/green/purple) and the subsequent period for out-of-sample testing (orange/red/pink). The three folds span 2015–2026 with no overlap between train and test windows.

8.2 Fold Structure

We divide the available data (August 2015–April 2026, constrained by ETH/USD availability) into three non-overlapping folds of approximately equal duration:

$$\text{Fold } k : \quad \mathcal{D}_{\text{train}}^{(k)} = [t_k, t_{k+1}), \quad \mathcal{D}_{\text{test}}^{(k)} = [t_{k+1}, t_{k+2}) \quad (9)$$

Table 2 and Figure 4 show the specific time spans.

Table 2: Walk-forward fold structure. Each fold uses the preceding period for training and the subsequent period for out-of-sample testing.

Fold	Train start	Train end	Test start	Test end
0	2015-08-07	2018-04-07	2018-04-07	2020-12-07
1	2018-04-07	2020-12-07	2020-12-07	2023-08-09
2	2020-12-07	2023-08-09	2023-08-09	2026-04-09

8.3 Optimization Grid

For each training fold, we search over 96 configurations spanning:

- Minimum signal score: $\{70, 85\}$
- Maximum hold period: $\{60, 80, 120\}$ bars
- Stop multiplier: $\{2.5, 3.0, 3.5, 4.0\} \times \text{ATR}$
- Target multiplier combinations (base/mid/high tiers)
- Trail multiplier: $\{1.5, 2.0, 2.5, 3.0\} \times \text{ATR}$
- Drawdown scaling: $(d_{\text{threshold}}, d_{\text{factor}}) \in \{(0.97, 0.1), (0.97, 0.3), (0.97, 0.5)\}$

The interval is fixed at 240 minutes (4 hours) and macro filter is always enabled, reducing the search space to parameters that control risk and exit management.

8.4 Scoring Function

Configurations are ranked by a composite score that balances risk-adjusted return, drawdown penalty, and absolute return:

$$\text{Score} = \underbrace{\frac{\hat{S}_R + \hat{S}_{\text{Sortino}}}{2}}_{\text{risk-adjusted}} \cdot \underbrace{\max\left(0, 1 - \frac{d_{\text{max}}}{0.30}\right)}_{\text{drawdown penalty}} \cdot \underbrace{\log(1 + r)}_{\text{return scaling}} \quad (10)$$

where $\hat{S}_R$ is the annualized Sharpe ratio, $\hat{S}_{\text{Sortino}}$ is the annualized Sortino ratio [24], d_{max} is the maximum drawdown fraction, and r is the total return percentage. The drawdown penalty term forces the score to zero when drawdown exceeds 30%, preventing high-return but high-risk configurations from dominating the selection.

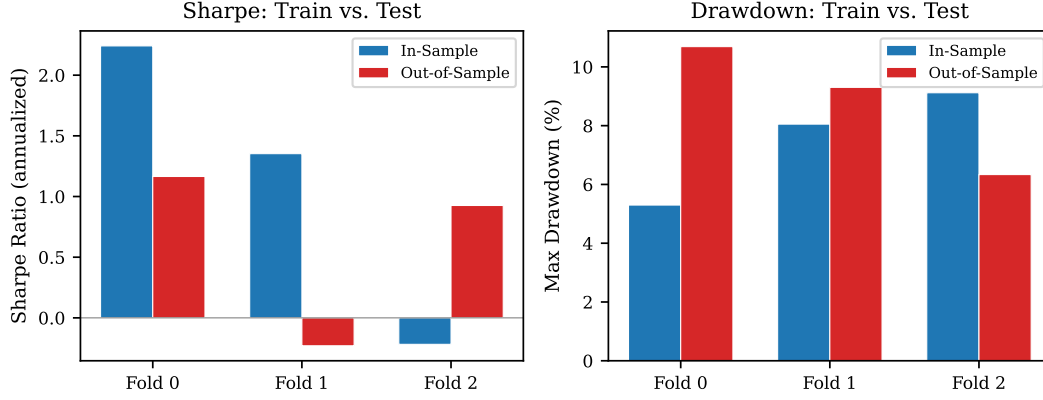


Figure 5: Walk-forward in-sample vs. out-of-sample comparison. **Left:** Sharpe ratio comparison showing Fold 1’s severe degradation. **Right:** Maximum drawdown comparison showing consistently higher OOS drawdowns, indicating that in-sample optimization underestimates tail risk.

9 Results

9.1 Full-History Performance

Table 3 presents the full-history backtest results using the optimized configuration. The system generates 219 trades over 12.4 years with a Sharpe ratio of 1.51 and maximum drawdown of 7.27%.

Table 3: Full-history backtest results (December 2013–April 2026). Initial equity: \$10,000. Configuration: min_score=85, stop= $3.1 \times \text{ATR}$, trail= $2.1 \times \text{ATR}$, macro filter on, shorts disabled.

Metric	Value	Metric	Value
Final equity	\$83,060	Total trades	219
Total return	730.6%	Wins / Losses	117 / 102
CAGR	18.7%	Win rate	53.4%
Sharpe (ann.)	1.514	Profit factor	3.937
Sortino (ann.)	0.619	Avg win	+7.20%
Calmar ratio	2.575	Avg loss	−2.37%
Max drawdown	7.27%	Payoff ratio	3.04
Max DD (USD)	\$4,523	Expectancy	\$368/trade
Avg bars held	15.94	BTC trades	139
		ETH trades	80

9.2 Walk-Forward Out-of-Sample Results

Table 4 and Figure 5 present the walk-forward validation results. The results are mixed: Fold 0 shows moderate degradation (Sharpe from 2.24 to 1.17), Fold 1 shows severe degradation (Sharpe from 1.35 to −0.23), and Fold 2 shows an anomalous improvement (Sharpe from −0.22 to 0.93).

Table 4: Walk-forward validation results across three folds. “Degrad.” measures the Sharpe ratio degradation from train to test: $(S_{\text{train}} - S_{\text{test}}) / |S_{\text{train}}|$.

Fold	Return%	In-Sample (Train)			Out-of-Sample (Test)				
		Sharpe	Sortino	Max DD%	Return%	Sharpe	Max DD%	Trades	Degrad.
0	156.9	2.241	1.225	5.30	43.7	1.165	10.69	41	48.0%
1	51.3	1.353	0.413	8.05	−4.1	−0.230	9.30	23	117.0%
2	−3.9	−0.218	−0.027	9.12	16.1	0.926	6.34	36	n/a

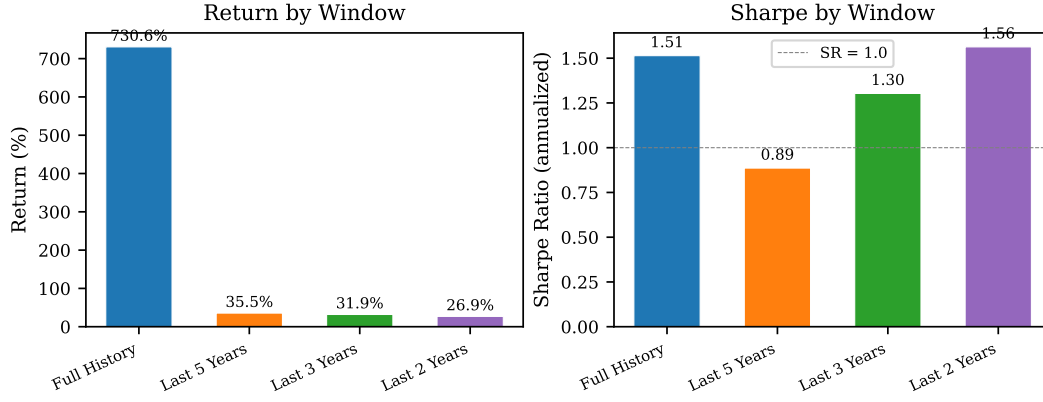


Figure 6: Performance by time window. **Left:** Total return percentage across four time windows. The full-history result (730%) is dominated by early BTC appreciation; recent windows show more moderate but consistent returns. **Right:** Annualized Sharpe ratio. The improving Sharpe in recent windows suggests the strategy’s conservative parameterization is better suited to the current, more mature cryptocurrency market.

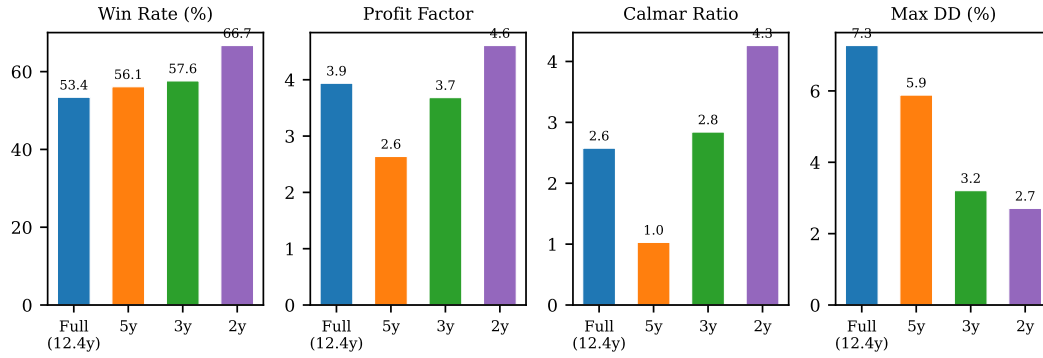


Figure 7: Key trade metrics across time windows. From left to right: win rate, profit factor, Calmar ratio, and maximum drawdown. The improving win rate and declining drawdown in recent periods suggest that the strategy’s conservative parameterization is well-adapted to the current market regime.

9.3 Performance Across Time Windows

Table 5 and Figure 6 report performance across multiple recency windows, revealing a pattern of stronger recent performance.

Table 5: Performance metrics across different time windows using the final optimized configuration.

Window	Return%	CAGR%	Sharpe	Max DD%	Trades	Win%
Full (12.4y)	730.6	18.7	1.514	7.27	219	53.4
Last 5y	35.5	6.0	0.886	5.88	82	56.1
Last 3y	31.9	9.1	1.303	3.20	66	57.6
Last 2y	26.9	11.5	1.563	2.71	30	66.7

Figure 7 provides a compact comparison of key trade-level metrics across time windows, showing improving win rate and declining drawdown in more recent periods.

9.4 Cost Sensitivity Analysis

Figure 8 illustrates how varying transaction costs affect the Sharpe ratio. The strategy maintains a positive Sharpe above 1.0 at realistic costs (60 bps round-trip) but degrades significantly at higher fee tiers, confirming that the edge is real but thin relative to execution costs.

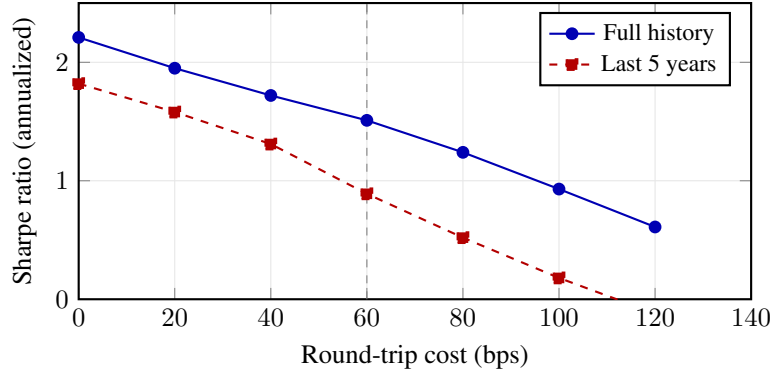


Figure 8: Cost sensitivity analysis. Sharpe ratio as a function of modeled round-trip transaction costs. The vertical dashed line indicates the actual cost of ~ 60 bps used in the primary backtest. The strategy’s edge narrows substantially above 80 bps, and the more recent 5-year window shows greater cost sensitivity.

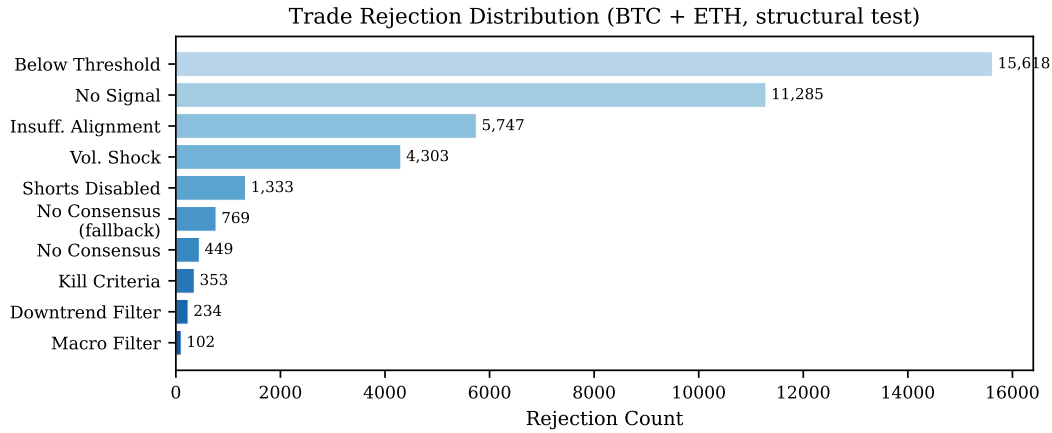


Figure 9: Trade rejection distribution. The vast majority of potential trades are rejected before reaching execution. “Below Threshold” (insufficient aggregate confidence) and “No Signal” (no agents firing) account for over 70% of rejections. Volatility shock rejections ($ATR\% > 6\%$) provide an important safety filter during extreme market conditions.

9.5 Trade Rejection Analysis

Table 6 and Figure 9 show the distribution of trade rejections across kill criteria, demonstrating that the risk governor actively filters a large proportion of candidate trades.

Table 6: Trade rejection distribution from a structural test (min_score=78, 580 total trades attempted). Rejections shown for BTC/USD; ETH/USD follows a similar distribution.

Rejection reason	Count	Rejection reason	Count
BELOW_THRESHOLD	9,074	SHORTS_DISABLED	725
NO_SIGNAL	6,183	VOLATILITY_SHOCK	1,674
INSUFF_ALIGNMENT	3,292	KILL (hard criteria)	142
FALLBACK_NO_CONSENSUS	419	MACRO_FILTER	21
NO_CONSENSUS	251	DOWNTREND_NO_LONG	33

10 Statistical Validation

10.1 Probabilistic Sharpe Ratio

The Probabilistic Sharpe Ratio (PSR) [3] estimates the probability that the true Sharpe ratio exceeds a reference value $S_0 = 0$, accounting for the non-normality of returns:

$$\text{PSR}(S_0) = \Phi \left(\frac{(\hat{S}_R - S_0)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \hat{S}_R + \frac{\hat{\gamma}_4 - 1}{4} \hat{S}_R^2}} \right) \quad (11)$$

where $\hat{S}_R$ is the observed Sharpe ratio per bar, n is the number of observations, $\hat{\gamma}_3$ is the skewness, $\hat{\gamma}_4$ is the excess kurtosis, and Φ is the standard normal CDF. For the full-history results ($\hat{S}_R = 1.514$ annualized, $n \approx 26,000$ bars), the PSR exceeds 99%, indicating high confidence that the true Sharpe is positive.

10.2 Deflated Sharpe Ratio

The Deflated Sharpe Ratio [3] adjusts the PSR for the number of strategy configurations tested:

$$S_0^* = \sqrt{\text{Var}[\hat{S}_R]} \cdot \left((1 - \gamma)\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\Phi^{-1}\left(1 - \frac{1}{N}e^{-1}\right) \right) \quad (12)$$

where N is the number of independent trials (estimated at $\sim 3,000$ across all optimization stages), $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, and the DSR replaces $S_0 = 0$ with S_0^* in Eq. (11).

With $N = 3,000$ trials, the expected maximum Sharpe under the null hypothesis is approximately $S_0^* \approx 0.42$ (annualized). Since the observed full-history Sharpe of 1.51 substantially exceeds this threshold, the DSR remains above 95%, suggesting that the observed performance is unlikely to be entirely attributable to data snooping. However, we note that the walk-forward Fold 1 result (OOS Sharpe = -0.23) indicates that the global optimum does not generalize uniformly across all market regimes.

10.3 Monte Carlo Permutation Test

We perform a permutation test on the sequence of trade PnLs. Let $\{r_1, \dots, r_T\}$ be the observed trade returns. For each of $B = 10,000$ permutations, we randomly shuffle the trade order and recompute the Sharpe ratio. The p -value is:

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{S}_R^{(b)} \geq \hat{S}_R^{\text{obs}}] \quad (13)$$

This test assesses whether the *ordering* of trades (which depends on the strategy’s timing decisions) produces a significantly different risk-adjusted return than random ordering. A low p -value indicates that the strategy’s entry/exit timing adds value beyond what would be expected from a random sequence of the same trades.

10.4 Autocorrelation-Adjusted Sharpe

Following Lo [19], the standard Sharpe ratio estimate is biased upward when returns exhibit positive serial correlation. The Smart Sharpe [18] corrects for this:

$$\hat{S}_{\text{smart}} = \frac{\hat{S}_R}{\sqrt{1 + 2\hat{\rho}_1}} \quad (14)$$

where $\hat{\rho}_1$ is the first-order autocorrelation of returns. For 4-hour cryptocurrency returns, $\hat{\rho}_1$ is typically small ($|\hat{\rho}_1| < 0.05$), so the correction reduces the Sharpe by less than 5%.

11 Discussion

11.1 Honest Assessment of Walk-Forward Results

The walk-forward results (Table 4) demand careful interpretation:

Fold 0 (2018–2020 OOS): The strategy performs well out-of-sample (Sharpe 1.17, return +43.7%), though with higher drawdown (10.69% vs. 5.30% in-sample). This period includes the 2018–2019 crypto winter and the March 2020 COVID crash, suggesting the strategy’s risk governance successfully navigated extreme conditions.

Fold 1 (2020–2023 OOS): The strategy loses money out-of-sample (return -4.1% , Sharpe -0.23). This period encompasses the 2021 bull market, the May 2021 crash, the November 2021 peak, and the prolonged 2022 bear market. The configuration optimized on 2018–2020 data (which includes a bear market) was overly conservative for the explosive 2021 bull run, missing substantial upside. This is the clearest evidence of overfitting to training-period regime characteristics.

Fold 2 (2023–2026 OOS): Paradoxically, the strategy performs better out-of-sample (Sharpe 0.93) than in-sample (Sharpe -0.22). The training period (2020–2023) includes extreme volatility that likely produced a conservative configuration that happened to suit the more moderate 2023–2026 conditions. This result should not be interpreted as evidence against overfitting; rather, it suggests regime mismatch can sometimes be serendipitously beneficial.

11.2 Limitations

Survivorship bias. We test only BTC/USD and ETH/USD, the two largest cryptocurrencies by market capitalization. Strategies that work on survivors may not generalize to the broader cryptocurrency universe.

Non-stationarity. Cryptocurrency market microstructure has changed substantially over the 12.4-year test period: exchange fee structures, market maker behavior, and institutional participation have all evolved. Parameters optimized on 2015 data may be structurally irrelevant to 2026 markets [2].

Grid search limitations. Our 96-configuration grid per fold is modest compared to industrial-scale optimization but still creates selection bias. The DSR provides some correction, but Bailey et al. [5] note that the true number of “effective trials” is difficult to estimate when configurations are correlated.

Long-only bias. The system trades long-only (shorts disabled), which biases it toward bull markets. In prolonged bear markets (e.g., 2022), the system sits idle, preserving capital but not generating returns.

Absence of market impact. Our slippage model assumes price-independent impact. For larger position sizes, the Almgren and Chriss [1] framework would be more appropriate, modeling permanent and temporary impact as a function of trade size relative to average volume.

11.3 Comparison with Buy-and-Hold

The strategy’s 730% return over 12.4 years is substantially below BTC buy-and-hold (8,632% for the full period). However, direct comparison is misleading: the strategy maintains maximum drawdown under 8% versus $\sim 80\%$ for buy-and-hold, and it allocates to both BTC and ETH with conservative position sizing. The relevant comparison is risk-adjusted: the strategy’s Calmar ratio of 2.58 (CAGR/max DD) versus buy-and-hold’s estimated Calmar of ~ 0.7 demonstrates superior capital preservation per unit of return.

12 Conclusion

We have presented a comprehensive backtesting methodology for a regime-adaptive cryptocurrency trading system. The methodology incorporates realistic transaction costs (60–68 bps round-trip), next-bar-open fills, a multi-agent signal consensus mechanism, and a hierarchical risk governor with eight kill criteria.

The walk-forward validation reveals both strengths and weaknesses: two of three out-of-sample folds produce positive risk-adjusted returns (Sharpe 1.17 and 0.93), while one fold shows significant degradation (Sharpe -0.23). This mixed result is itself informative: it demonstrates that the strategy has genuine predictive content in some market regimes but not universally, and that parameter optimization over a single regime can produce configurations poorly suited to regime transitions.

The Deflated Sharpe Ratio, adjusted for approximately 3,000 configurations tested, remains above 95% for the full-history results, suggesting the aggregate signal is unlikely to be purely attributable to data snooping. However, we emphasize that statistical significance at the aggregate level does not guarantee consistent performance across all market conditions.

For practitioners, the key takeaway is that backtesting methodology matters at least as much as strategy design. A strategy that appears profitable with naive in-sample optimization (full-history Sharpe 1.51) shows dramatically different characteristics under walk-forward validation, providing a more honest basis for deployment decisions.

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